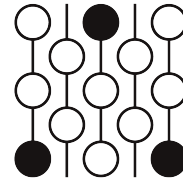


Review of Submission 2B

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. The AI got the problem correct. This is a working proof, with all essential details included and perhaps even some extra details. I especially appreciated the excellent proof overview at the top of the solution.

The proof in the first half works well, and it follows a clear outline. All details seem correct. Sometimes, as in Lemma 5 and 6, the proof did not seem to be the most efficient or transparent.

The proof in the second half seems to work, though I found the exposition somewhat hard to follow. The AI supplies lots of very fine-scale details for things which fairly obviously work. To give you a feel for this, if you have a smooth compact embedded loop in the plane, it is clear that a sufficiently fine polygonal approximation, with points on the loop, will be embedded. If you include all kinds of estimates for this, it mostly clouds the picture. Something similar is going on here, one dimension up.

Novelty. Let me first discuss the context for the problem and then describe how the AI did. The problem is based on my solution of the Halpern-Weaver Conjecture, a result which asks for the infimal aspect ratio needed for a smooth and isometric embedding of a flat Moebius band into $\mathbb{R}^3$. My paper was published in the *Annals* in January, 2025 and there are also some of accounts of the result in the popular media. The problem the AI considered is the same problem in a different category of maps, piecewise linear maps with a certain side condition called squeezing.

The first half of the proof really gave me vertigo! The AI supplied essentially all the arguments in my paper, though always in a kind of disguised or transmuted form. For instance, the two functions used in Lemma 6 are equivalent to the ones used in my paper, but they are presented in very different notation and you need an auxiliary vector identity to see the equivalence. For a while I wondered whether the AI had (amazingly) come up with its own proof without looking at my paper, but then I noticed that it used the same letters T, B, D, H for the four main objects in the proof. (I used these letters because T stands for “top” and B stands for “bottom” and D stands for “diagonal” and H stands for “horizontal”. The AI used these specific letters without explanation.) This was the only “tell” that the AI had read my paper. Once I realized this, I was annoyed. So, the first half is novel in some fine-scale details but overall derived from my paper.

The second half is not derived from my paper, and it essentially gives a nuts-and-bolts proof involving smooth approximation of an already known smooth construction that goes back to Michael

Sadowsky in 1930. I didn't really like this proof that much. It uses a black box, Sadowsky's result, and then grinds away with PL approximation until it wins. So, this half is sort of novel, but it is what you would do if you did not really have any artistic sense.

2 Recommendation

I would recommend that this paper is publishable with moderate revisions. (If I have to decide between "major" and "minor" I would go with "major", but I am kind of on the fence about it.) The most important complaint I have is that the AI did not acknowledge my paper at all, even though it clearly read it. Minor complaints about the first half: As I mentioned above, I thought that Lemmas 5 and 6 were not that efficient. Lemma 5 uses calculus in a way that suppresses the main calculation; a swift linear algebra argument would be better here. Lemma 6 uses a very opaque form of my functions. A more geometrically transparent formulation of the second one would save the reader a lot of head scratching.

I think that the second half should be reworked. The reduction of the PL case to an already known result in the smooth case is certainly acceptable for a proof, though if you are an editor with some mathematical taste you would probably ask for a better argument. One way the argument could be better is if it gave an explicit and self-contained construction. The other way it would be better is if it did the PL approximation in a cleaner way, with the ideas emphasized rather than the calculations. (To be fair, the overview at the top of the solution does a good job of giving the high level strategy for this.)

The last thing I want to say is that I would be tempted to fail a student on an exam for clearly copying ideas from an external source without acknowledging this. In a human, this lack of idea-acknowledgement would seem like an ethical failing. I don't know what to say about how it reflects on an AI. But, anyway, if I was in a bad mood I might reject the paper on these kinds of grounds.

3 Submission with inline comments

Theorem 1 (Headline theorem: original problem). Let $\mathbb{Z}$ denote the integers and let $\mathbb{R}$ denote the reals. Let

$$\Sigma = \mathbb{R} \times [0, 1], \quad \partial\Sigma = \mathbb{R} \times \{0, 1\}.$$

For $\beta > 0$, define $G_\beta : \Sigma \rightarrow \Sigma$ by

$$G_\beta(x, y) = (x + \beta, 1 - y).$$

Let G_β^k denote the k -fold composition of G_β or G_β^{-1} , according as $k > 0$ or $k < 0$, and let G_β^0 be the identity. A *clean triangle* is the convex hull of three non-collinear points in $\partial\Sigma$. A *clean triangulation* of Σ is a countable, locally finite collection of clean triangles with pairwise disjoint interiors and union Σ . It is *G_β -invariant* if G_β preserves the triangulation. A *squeeze map* of a clean triangle is an affine map into $\mathbb{R}^3$ which decreases the length of the edge contained in $\partial\Sigma$ and increases the lengths of the other two edges. A number $\beta > 0$ is *realized* if there is a continuous map $f : \Sigma \rightarrow \mathbb{R}^3$ such that

1. f restricts to a squeeze map on each triangle of a G_β -invariant clean triangulation of Σ ;
2. $f(p) = f(q)$ if and only if $p = G_\beta^k(q)$ for some $k \in \mathbb{Z}$.

Then the infimum of the set of realized numbers is $\sqrt{3}$.

Theorem 2 (Established stronger claim). *The realized numbers are exactly the interval $(\sqrt{3}, \infty)$.*

Overview / Strategy

Reviewer 1

I really like this overview. It makes it clear that the proof is going in the right direction.

The lower bound passes to the quotient $M_\beta = \Sigma / \langle G_\beta \rangle$, a flat Möbius band with boundary length 2β . The map descends to an embedded piecewise-affine surface. Inside each clean triangle the cevians from the apex to the boundary edge are strictly lengthened. These cevians form a leaf graph; in the quotient the leaf graph is a circle, while upstairs it is a line, and the deck transformation reverses the orientation of the image support line. A topological lemma about paths of oriented lines then forces two image support lines to be perpendicular and intersecting. Boundary shortening and a Euclidean estimate give $\beta > \sqrt{3}$.

For the upper bound we use one classical external input: the Sadowsky–Wunderlich construction of embedded developable Möbius bands, in a quantitative form. We approximate such a smooth ruled band by an invariant piecewise-linear ruled band whose boundary edges are not expanded and whose transverse edges are shortened by an arbitrarily small amount. A final anisotropic rescaling makes the boundary edges strictly shorter and all transverse edges strictly longer, yielding a realization for every $\beta > \sqrt{3}$.

3.1 The lower bound

Throughout this section assume that β is realized by f on a G_β -invariant clean triangulation $\mathcal{T}$. Write $G = G_\beta$.

Lemma 3 (Face-to-face structure). *The triangulation is face-to-face: distinct edges have disjoint relative interiors, every transverse edge is incident to exactly two triangles, and every boundary edge is incident to exactly one triangle.*

Proof. A clean triangle has exactly one boundary edge: among its three boundary vertices, exactly two lie on one boundary component and the third lies on the other. The remaining two edges are transverse.

Two transverse edges cannot cross in the interior of Σ , since near a transverse crossing each of the two incident triangles occupies one open side of its edge, and two such open half-disks always overlap in an open sector. If two transverse edges are collinear and overlap in more than one point, then they lie on the same non-horizontal line; such a line meets $y = 0$ and $y = 1$ once each, so the two edges are the same geometric segment. Boundary edges cannot overlap in a nontrivial interval, because at an interior point of the overlap both adjacent triangles contain an

inward half-neighborhood of the boundary. Finally, the endpoint of a transverse edge cannot lie in the relative interior of a boundary edge: the triangle with the boundary edge contains a small inward half-disk there, while the triangle with the transverse edge contains a nonempty sector inside that half-disk. Hence distinct triangles meet only in common faces.

Let e be a transverse edge and z an interior point of e . A small disk around z is divided by e into two half-disks. At most one triangle occupies each side, by disjointness of interiors, and at least one triangle occupies each side, because the triangles cover Σ . Thus e is incident to exactly two triangles. The same half-disk argument at an interior point of a boundary edge gives exactly one incident triangle. $\square$

Lemma 4 (The finite quotient triangulation). *The quotient map f descends to an embedding $F : M_\beta \rightarrow \mathbb{R}^3$, where*

$$M_\beta = \Sigma / \langle G \rangle = [0, \beta] \times [0, 1] / (0, y) \sim (\beta, 1 - y),$$

and the quotient triangulation of M_β is finite and face-to-face.

Proof. Since $p = Gq$ implies $f(p) = f(q)$, the map descends to a continuous F . If $F([p]) = F([q])$, then $f(p) = f(q)$, so by realization $p = G^k q$, hence $[p] = [q]$. Thus F is injective; because M_β is compact and $\mathbb{R}^3$ is Hausdorff, F is an embedding.

The restriction of f to each triangle is injective. Indeed, if an affine restriction had rank < 2 , then arbitrarily close distinct points p, q in that triangle would satisfy $f(p) = f(q)$. Then $p = G^k q$ for some $k \neq 0$, impossible for sufficiently close p, q , since G^{2n} translates by $2n\beta$ and G^{2n+1} changes the x -coordinate by $(2n+1)\beta$, so every nontrivial G^k moves every point by distance at least β .

Therefore no closed triangle meets a nontrivial translate of itself. If $z \in \tau \cap G^k \tau$, write $z = G^k w$ with $w \in \tau$. Then $f(z) = f(w)$, and injectivity on τ gives $z = w$, contradicting freeness of the G -action unless $k = 0$. A compact fundamental rectangle, for instance $[0, 2\beta] \times [0, 1]$, meets only finitely many closed triangles by local finiteness and compactness, and every orbit of triangles has a representative meeting it. Hence the quotient triangulation is finite. Lemma 3 descends to the quotient. $\square$

Reviewer 1

Lemma 5 here works, but the proof seems needlessly complicated. For one thing, the derivative computation is not obvious. For another thing, it is not really necessary. Since you have a quadratic function you can deduce the sign of the second derivative from the asymptotics at either end. Also, the whole thing can be done easily with linear algebra.

Lemma 5 (Cevian lengthening). *Let $\Delta = \text{conv}\{a, r_0, r_1\}$ be a clean triangle whose boundary edge is $[r_0, r_1]$, and let $A : \Delta \rightarrow \mathbb{R}^3$ be a squeeze map. For $r(s) = (1-s)r_0 + sr_1$, every cevian $[a, r(s)]$ is strictly lengthened:*

$$|A(a) - A(r(s))| > |a - r(s)|, \quad 0 \leq s \leq 1.$$

Proof. Set $q(s) = |A(a) - A(r(s))|^2 - |a - r(s)|^2$. The two transverse edges are strictly lengthened, so $q(0) > 0$ and $q(1) > 0$. Since A is affine,

$$q''(s) = 2(|A(r_1) - A(r_0)|^2 - |r_1 - r_0|^2) < 0$$

because the boundary edge is strictly shortened. Thus q is strictly concave and lies above its chord: $q(s) \geq (1-s)q(0) + sq(1) > 0$. Taking square roots gives the claim. $\square$

Reviewer 1

Lemma 6 works, and it is very similar to what is in my paper about the optimal Moebius band. The choice of the second function b is really wierd. It would be much more transparent to define b in terms of the identity that is established lower down. (See next comment.) Also, this is essentially Lemma T in my paper.

Lemma 6 (Paths of oriented lines). *Let L_s , $0 \leq s \leq 1$, be a continuous path of oriented lines in $\mathbb{R}^3$. Suppose L_1 is the same unoriented line as L_0 , but with the opposite orientation. Then two distinct parameters give unoriented lines that are perpendicular and intersect.*

Proof. Represent an oriented line by (u, m) , where $u \in S^2$ is its unit direction and $m = p \times u$ for any point p on the line. Reversing orientation sends (u, m) to $(-u, -m)$. Choose continuous $(u(s), m(s))$ and extend by $(u(s+1), m(s+1)) = -(u(s), m(s))$. For $s \in \mathbb{R}/\mathbb{Z}$ and $0 \leq \delta \leq 1$, define

$$a(s, \delta) = u(s) \cdot u(s + \delta), \quad b(s, \delta) = u(s) \cdot m(s + \delta) + u(s + \delta) \cdot m(s).$$

Then $(a, b)(s, 0) = (1, 0)$, $(a, b)(s, 1) = (-1, 0)$, and $(a, b)(s + \delta, 1 - \delta) = -(a, b)(s, \delta)$.

Assume $(a, b) \neq (0, 0)$ everywhere. Then $\Phi = (a, b)/\sqrt{a^2 + b^2} : S^1 \times [0, 1] \rightarrow S^1$ is continuous and is constant on $\delta = 0$. On $\delta = 1/2$,

$$\Phi(s + 1/2, 1/2) = -\Phi(s, 1/2).$$

If $\psi : S^1 \rightarrow S^1$ has $\psi(s + 1/2) = -\psi(s)$, then, after lifting to an angle θ , the integer $(\theta(s + 1/2) - \theta(s) - \pi)/(2\pi)$ is constant, so the degree of ψ is odd. But $\Phi|_{\delta=1/2}$ is homotopic through $S^1 \times [0, 1/2]$ to the constant map $\Phi|_{\delta=0}$, so its degree is zero, a contradiction. Hence $(a, b) = (0, 0)$ somewhere, with $0 < \delta < 1$, so the two parameters are distinct.

At such a zero, $u \cdot v = 0$ for the two directions u, v . If the lines are represented by $(u, p \times u)$ and $(v, q \times v)$, then

Reviewer 1

It would be much more geometrically clear to define b in terms of the right side of this equality, and then explain why it does not depend on the choice of p, q .

$$b = u \cdot (q \times v) + v \cdot (p \times u) = (p - q) \cdot (u \times v).$$

Thus, when $u \perp v$, the condition $b = 0$ says the two nonparallel lines are coplanar. Nonparallel coplanar lines intersect. $\square$

Lemma 7 (Perpendicular intersecting leaves). *In M_β there are two distinct leaves whose image support lines under F are perpendicular and intersect.*

Proof. Inside each triangle take the cevians from the apex to the boundary edge and call them leaves. In the finite quotient triangulation, form a graph Λ_M by taking one closed interval for the leaves of

each triangle and gluing endpoints corresponding to common transverse edges. By Lemma 3, every vertex has valence 2.

The graph is connected. If C is a component and $U_C \subset M_\beta$ is the union of the corresponding leaves, then U_C is open: this is immediate inside triangles, along boundary edges, and across transverse edges because the two incident leaf intervals are glued. At a boundary vertex, a small half-disk meets finitely many incident triangle sectors; consecutive sectors share an edge from the vertex, and all non-boundary shared edges are transverse, so all incident triangles lie in the same component. Thus the sets U_C are disjoint open subsets covering the connected surface M_β ; hence there is only one. A finite connected graph with all vertices of valence 2 is a circle, so $\Lambda_M \cong S^1$.

The same openness argument upstairs gives a connected leaf graph Λ_Σ . The action of G on leaves is free. A lifted leaf has endpoints $(x, 0)$ and $(x+t, 1)$. If G^{2n} fixes it, then $2n\beta = 0$. If G^{2n+1} fixes it, then the two endpoint equations give simultaneously $t = (2n+1)\beta$ and $t = -(2n+1)\beta$, impossible. Hence $\Lambda_\Sigma \rightarrow \Lambda_M$ is a connected infinite covering of a circle, so $\Lambda_\Sigma \cong \mathbb{R}$.

Orient every lifted leaf from $y = 0$ to $y = 1$. By Lemma 5, its image is a nondegenerate segment, hence determines a continuous oriented support line along Λ_Σ . If ℓ is a leaf, then $G\ell$, reoriented from bottom to top, has the same image segment with the opposite orientation, because G swaps boundary components and $f \circ G = f$. Applying Lemma 6 to the compact interval in Λ_Σ from ℓ to $G\ell$ gives two distinct quotient leaves with perpendicular intersecting support lines. $\square$

Reviewer 1

This is essentially a copy of the main geometric idea in my paper on the optimal paper Moebius band.

Reviewer 1

A diagram explaining the above construction would be extremely helpful. Of course, the proof works without the diagram, but it is much harder to follow without it.

Proposition 8 (No $\beta \leq \sqrt{3}$ is realized). *If β is realized, then $\beta > \sqrt{3}$.*

Proof. Let J be a lifted leaf with endpoints $(x, 0)$ and $(x+t, 1)$. Its flat length is $\sqrt{1+t^2}$. We first show

$$|t| < \beta. \quad (1)$$

If $t = \beta$ or $t = -\beta$, the endpoints are identified by G or G^{-1} , so their images coincide, contradicting Lemma 5. If $t > \beta$, translate so J has endpoints $(0, 0)$, $(t, 1)$. Then GJ has endpoints $(\beta, 1)$, $(t+\beta, 0)$, and the two segments meet at height $(t+\beta)/(2t) \in (0, 1)$. Thus some distinct $p, q \in J$ satisfy $p = Gq$, so $f(p) = f(q)$, contradicting injectivity on the nondegenerate image segment. The case $t < -\beta$ is identical using G^{-1} .

Parametrize the boundary circle of M_β modulo 2β by assigning coordinate u to $(u, 0)$ and coordinate $v - \beta$ to $(v, 1)$. Hence the boundary arcs between the endpoints of a leaf of span t have lengths

$$\beta - t, \quad \beta + t, \quad (2)$$

both positive by (1). Also every nonconstant boundary arc is strictly shortened by F : the quotient boundary is subdivided into finitely many boundary edges, and on each such edge F has constant

length factor $\lambda_e < 1$; summing over subsegments gives

$$\ell_F(\gamma) < \ell_0(\gamma). \quad (3)$$

Reviewer 1

Aha! Here T and B are in my paper. In my paper, T stands for “top” and B stands for “bottom”. This is when I figured out that the AI read my paper and used the ideas without attribution.

Choose the two leaves T, B from Lemma 7, and let their support lines meet perpendicularly at P . Since distinct leaves have disjoint relative interiors and F is embedded, P cannot lie in the relative interiors of both image segments. Rename so that $F(B)$ lies in a closed ray from P . Let q be the endpoint of B farther from P , and put $Q = F(q)$. Then

$$y := \text{dist}(Q, \text{line } F(T)) = |Q - P| \geq |F(B)| > |B|_0 \geq 1,$$

so in fact

$$y > 1. \quad (4)$$

Reviewer 1

The letter D for this boundary arc also appears in my paper. Here D stands for “diagonal” in my paper, based on the geometry of the situation. The same comments apply to the letter H , which stands for “horizontal”.

Lift T with endpoints $(x, 0), (x + t, 1)$. Replacing this lift by its G -translate changes t to $-t$, so it interchanges the two lengths in (2). Choose the lift so that the boundary arc D containing q has length $\beta + t$; let the other arc be H , of length $\beta - t$. Let A, C be the endpoints of $F(T)$, and set

$$L = |A - C| = |F(T)| > \sqrt{1 + t^2}. \quad (5)$$

The arc H connects A to C , so by (3),

$$\beta - t = \ell_0(H) > \ell_F(H) \geq L > \sqrt{1 + t^2},$$

hence

$$\beta > \sqrt{1 + t^2} + t. \quad (6)$$

The arc D connects A to C and passes through Q . Therefore $\ell_F(D) \geq |A - Q| + |Q - C|$. Among points at distance y from the line AC , the sum of distances to A and C is minimized when the projection is the midpoint of AC , giving

$$|A - Q| + |Q - C| \geq \sqrt{L^2 + 4y^2}.$$

Using (4) and (5),

$$\ell_F(D) > \sqrt{5 + t^2}.$$

By (3),

$$\beta + t = \ell_0(D) > \ell_F(D) > \sqrt{5 + t^2},$$

so

$$\beta > \sqrt{5+t^2} - t. \quad (7)$$

Combining (6) and (7),

$$\beta > \max\{\sqrt{1+t^2} + t, \sqrt{5+t^2} - t\}.$$

For every real t , this maximum is at least $\sqrt{3}$. Indeed, if both terms were $< \sqrt{3}$, then squaring $\sqrt{1+t^2} + t < \sqrt{3}$ gives $t < 1/\sqrt{3}$, while squaring $\sqrt{5+t^2} - t < \sqrt{3}$ gives $t > 1/\sqrt{3}$, a contradiction. Hence $\beta > \sqrt{3}$. $\square$

3.2 The upper bound

Reviewer 1

Basing the proof on the Sadowski construction works, but it might be cleaner to base the proof on the construction in the 1977 Halpern-Weaver paper, which gives a C^∞ example. Then you don't have to worry as much about how the regularity plays with the approximation. For whatever is used here, I'm not crazy about the fact that the proof uses a black box here.

We quote the classical Sadowsky–Wunderlich construction in the precise form needed.

Theorem 9 (Sadowsky–Wunderlich input). *For every $\mu > \sqrt{3}$, there is a map $E_\mu : \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}^3$ such that:*

1. $E_\mu(X + \mu, 1 - Y) = E_\mu(X, Y)$;
2. *the induced quotient Möbius band is embedded;*
3. *the quotient surface is $C^{1,1}$, piecewise C^∞ , and developable;*
4. *it is foliated by full-width straight rulings, and E_μ maps each ruling affinely and isometrically to a segment;*
5. *the boundary lifts are parametrized by arclength in the X -coordinate;*
6. *the construction remains embedded on a slightly wider strip.*

Proof. This is the three-cylinder construction of Sadowsky, with Wunderlich's analysis of the developable Möbius band; see [1, 2]. In that construction the band consists of three planar pieces and three cylindrical pieces joined along generators with matching tangent planes, and the bending is isometric. The standard separation computation gives maximum half-width

Reviewer 1

How is this complicated calculation relevant? It would be better to just explain qualitatively what is going on.

$$b_{\max} = \frac{\mu}{2\sqrt{3}} - \frac{3\pi d}{4}$$

for auxiliary cylinder diameter $d > 0$. Since $\mu > \sqrt{3}$, choosing $0 < d < 2(\mu - \sqrt{3})/(3\pi\sqrt{3})$ gives $b_{\max} > 1/2$, hence a width-one embedded strip, and the strict inequality gives the slightly wider embedded extension. $\square$

For $L, c > 0$, write

$$H_{L,c}(X, Y) = (X + L, c - Y) \quad \text{on } \mathbb{R} \times [0, c].$$

Lemma 10 (Invariant PL approximation). *Let $E : \mathbb{R} \times [0, c] \rightarrow \mathbb{R}^3$ be a scaled Sadowsky band invariant under $H_{L,c}$, with full-width rulings*

$$R_x = [(x, 0), (h(x), c)].$$

Fix x_0 , put $\phi(x) = h(x) + L$, and set

$$S = \max_{x \in [x_0, \phi(x_0)]} |h(x) - x|.$$

For every $D > S$ and every $\varepsilon > 0$, there are an $H_{L,c}$ -invariant clean triangulation $\mathcal{T}_0$ and an $H_{L,c}$ -invariant PL map F_0 such that:

1. $F_0(P) = F_0(Q)$ iff $P = H_{L,c}^k(Q)$ for some $k \in \mathbb{Z}$;
2. every boundary edge is nonexpanded;
3. every transverse edge PQ satisfies

$$|F_0(P) - F_0(Q)| \geq |P - Q| - \varepsilon;$$

4. every transverse edge has horizontal span $< D$.

Proof. The noncrossing full-width rulings imply that h is a strictly increasing homeomorphism. The map $H = H_{L,c}$ sends R_x to $R_{\phi(x)}$, so

$$h(\phi(x)) = x + L, \quad \phi^2(x) = x + 2L. \tag{8}$$

Also

$$|h(x) - x| < L. \tag{9}$$

If $h(x) - x = L$, the endpoints of R_x are identified by H , contradicting the isometric nondegenerate image of R_x . If $h(x) - x > L$, then R_x meets $H(R_x)$ in the interior of the strip; hence two distinct points of R_x have the same image under E , again impossible. The case $h(x) - x < -L$ follows using H^{-1} . Thus $\phi(x) > x$.

Choose $\rho > 0$ with $S + \rho < D$ and $2\rho < \varepsilon$. Choose a partition

$$x_0 < x_1 < \cdots < x_N = \phi(x_0)$$

so fine that $x_{i+1} - x_i < \rho$, $h(x_{i+1}) - h(x_i) < \rho$, all seams of the piecewise smooth Sadowsky band occur among the rulings R_{x_i} , and the embeddedness condition proved below holds. Extend by $x_{i+N} = \phi(x_i)$. Then $x_{i+2N} = x_i + 2L$. Let

$$B_i = (x_i, 0), \quad T_i = (h(x_i), c).$$

The quadrilaterals $B_i B_{i+1} T_{i+1} T_i$ tile the strip. For $0 \leq i < N$, triangulate by the diagonal $B_i T_{i+1}$, and extend H -invariantly. Define F_0 on vertices by $F_0(B_i) = E(B_i)$, $F_0(T_i) = E(T_i)$, and extend affinely on triangles. Then F_0 is H -invariant.

Boundary edges are nonexpanded because their images are chords of boundary arcs parametrized by arclength. A ruling edge is mapped isometrically. For a diagonal $B_i T_{i+1}$, set $d = |B_i - T_{i+1}|$ and $r = |B_i - T_i|$. Since $|T_i - T_{i+1}| < \rho$, one has $|d - r| < \rho$, and

$$|F_0(B_i) - F_0(T_{i+1})| \geq |E(B_i) - E(T_i)| - |E(T_i) - E(T_{i+1})| > r - \rho > d - 2\rho > d - \varepsilon.$$

The other diagonal type is identical, using $x_{i+1} - x_i < \rho$. By (8), the absolute span $|h(x) - x|$ is preserved under ϕ , so all ruling spans are at most S ; diagonal spans are $< S + \rho < D$.

Reviewer 1

Here is where using a smooth example would be nicer than $C^{1,1}$. The reader wouldn't have think as hard about the convergence.

It remains to justify the promised embeddedness for sufficiently fine partition. Let $N = (\mathbb{R} \times [0, c]) / \langle H \rangle$. By Theorem 9, E extends to a slightly wider compact embedded $C^{1,1}$ surface $\widehat{N}$, so there is a tubular neighborhood U with nearest-point projection $\pi : U \rightarrow \widehat{E}(\widehat{N})$. The secant PL maps converge uniformly to E ; because E is affine on rulings, $C^{1,1}$ on each piece, and the seams are included among the ruling edges with matching tangent planes, the maps

$$\Psi = \widehat{E}^{-1} \circ \pi \circ F_0 : N \rightarrow \widehat{N}$$

converge to the inclusion $N \hookrightarrow \widehat{N}$ in C^1 on each piece.

For sufficiently small C^1 -error, Ψ is injective. Choose finitely many coordinate disks with convex coordinate images and $\delta > 0$ so that points of N at distance $< \delta$ lie in a common disk. In such a disk, if the derivative of Ψ differs from the identity by $< 1/2$, then integration along the coordinate segment gives $|\Psi(p) - \Psi(q)| \geq \frac{1}{2}|p - q|$. For pairs with distance $\geq \delta$, compactness and injectivity of the inclusion give a positive minimum separation; sufficiently small C^0 -error preserves this separation. Hence Ψ is injective. If the quotient map induced by F_0 identified two points, then Ψ would identify them, contradiction. Thus the quotient PL map is embedded, which is equivalent to assertion (1) after lifting. $\square$

Proposition 11 (Every $\beta > \sqrt{3}$ is realized). *For every $\beta > \sqrt{3}$, there is a realization of β .*

Proof.

The definition of a here is rather complicated. It would be nicer just to explain conceptually why sufficiently small a works. The proof is already so nonconstructive that these fine scale details are not that helpful. Similar remarks apply to the other fine-scale calculations. The one main advantage of giving an approximation argument like this is that (done well) it would spare you the nitty-gritty details.

Choose μ with $\sqrt{3} < \mu < \beta$, and take the Sadowsky band E_μ . Let $S_\mu = \max |h(x) - x|$ over one fundamental interval of rulings, and choose $D_\mu > S_\mu$. Choose $a < 1$, close to 1, with $a > \mu/\beta$, and set

$$c = \frac{a\beta}{\mu}, \quad L = a\beta.$$

Then $c > 1$ and $L = c\mu$. Scale E_μ by c , giving a width- c , length- L band invariant under $H_{L,c}$. Its span bound scales to cS_μ .

Choose a close enough to 1 so that

$$m := \min_{0 \leq u \leq D_\mu} \left(c\sqrt{1+u^2} - \sqrt{\left(\frac{cu}{a}\right)^2 + 1} \right) > 0. \quad (10)$$

This is possible because at $a = 1$ one has $c = \beta/\mu > 1$, and $c\sqrt{1+u^2} > \sqrt{c^2u^2 + 1}$ on the compact interval $[0, D_\mu]$. Choose $0 < \varepsilon < m$ and apply Lemma 10 to the scaled band with $D = cD_\mu$. We get $\mathcal{T}_0$ and F_0 .

Define

$$A : \Sigma \rightarrow \mathbb{R} \times [0, c], \quad A(x, y) = (ax, cy).$$

Because $L = a\beta$,

$$A(G_\beta(x, y)) = (ax + a\beta, c - cy) = H_{L,c}(A(x, y)).$$

Thus A conjugates G_β to $H_{L,c}$. Put

$$f = F_0 \circ A, \quad \mathcal{T} = A^{-1}(\mathcal{T}_0).$$

Then $\mathcal{T}$ is a countable locally finite G_β -invariant clean triangulation, and f is affine on every triangle.

If pq is a boundary edge, then

$$|f(p) - f(q)| \leq |Ap - Aq| = a|p - q| < |p - q|.$$

If pq is a transverse edge, let s be the horizontal span of $ApAq$. By Lemma 10, $s < cD_\mu$; write $s = cu$ with $0 \leq u < D_\mu$. Then

$$|Ap - Aq| = c\sqrt{1+u^2}, \quad |p - q| = \sqrt{\left(\frac{cu}{a}\right)^2 + 1}.$$

Using Lemma 10 and $\varepsilon < m$, (10) gives

$$|f(p) - f(q)| \geq |Ap - Aq| - \varepsilon > |p - q|.$$

Thus f is a squeeze map on every triangle. Finally, for all $p, q \in \Sigma$,

$$f(p) = f(q) \iff F_0(Ap) = F_0(Aq) \iff Ap = H_{L,c}^k(Aq) \text{ for some } k \in \mathbb{Z} \iff p = G_\beta^k(q) \text{ for some } k \in \mathbb{Z}.$$

Therefore β is realized. $\square$

Proof of Theorem 2. Proposition 8 shows that every realized number is $> \sqrt{3}$. Proposition 11 shows that every number $> \sqrt{3}$ is realized. Hence the realized numbers are exactly $(\sqrt{3}, \infty)$. $\square$

Proof of Theorem 1. By Theorem 2, the set of realized numbers is $(\sqrt{3}, \infty)$, whose infimum is $\sqrt{3}$. $\square$

Reviewer 1

My paper on the optimal paper Moebius band should be cited. Given that these other papers are cited, it is pretty shocking that mine is not.

References

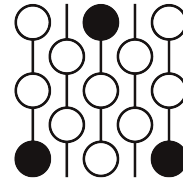
- [1] M. Sadowsky, “Ein elementarer Beweis für die Existenz eines abwickelbaren Möbiusschen Bandes und die Zurückführung des geometrischen Problems auf ein Variationsproblem,” *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse*, 1930, 412–415.
- [2] W. Wunderlich, “Über ein abwickelbares Möbiusband,” *Monatshefte für Mathematik* **66** (1962), 276–289.

Review of Submission 2B

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The solution provides a correct argument both for the lower bound $\beta \geq \sqrt{3}$ and for the realizability of every $\beta > \sqrt{3}$. It accomplishes the first goal by adapting Schwartz’s argument, which shows that a smoothly isometrically embedded Moebius strip has aspect ratio greater than $\sqrt{3}$, to the PL setting in which the triangles in the triangulation satisfy the “squeeze map” hypothesis. This adaptation is carried out without any major logical flaws.

For the second goal, the solution starts with Sadowsky’s family of C^1 embeddings of the flat Mobius strip, which have straight-line rulings and isometrically embedded boundary curves. It then approximates these embeddings by PL embeddings, which do not yet satisfy the squeeze property, and chooses scaling factors for the domain Mobius strip so that the resulting PL embedding does satisfy the squeeze property. The solution supports this part of the argument by justifying the choice of scaling constants and the size of the partition used in the PL approximation, as well as by verifying that the chosen scaling constants achieve the squeeze property on each triangle. However, the solution is unclear in its proof that the constructed PL embeddings are indeed embedded; see the inline comments for details. Nevertheless, I believe that the statement is still correct and can be salvaged by working directly with Sadowsky’s construction.

Novelty. For both parts of the argument, the strategies implemented by the solution are essentially the same as the expected strategy for this problem, the strategy followed by the human solution, and the strategies pursued by some of the other AI models. Therefore, this approach should not be considered novel.

For the first part, the solution identifies that the proof strategy underlying Schwartz’s theorem can be modified to prove the lower bound $\beta > \sqrt{3}$ in the setting of the problem statement. However, given access to Schwartz’s result, adapting the argument to the PL category is a relatively routine task. In particular, Lemmas 6, 7, and 8 use proof strategies that are very similar to those of corresponding lemmas in Schwartz’s paper. The solution however fails to cite Schwartz’s work explicitly in this part of the argument.

For the second part, the solution uses PL approximations of a known construction in the smooth setting. This is also a standard strategy, provided the necessary background from the existing literature is available.

Exposition. The solution gives lengthy proofs of some basic properties of the triangulation described in the problem statement; see the inline comments for Lemma 3 and 4. I believe that this part of the argument could be abbreviated. This issue does not affect the correctness of the argument.

2 Recommendation

I recommend that the submission be accepted after minor revisions. The main revisions needed are to give proper attribution to Schwartz's theorem in the proofs of Lemmas 6, 7, and 8, and to clarify the confusion in Lemma 10. I consider these issues correctable.

3 Submission with inline comments

Theorem 1 (Headline theorem: original problem). Let $\mathbb{Z}$ denote the integers and let $\mathbb{R}$ denote the reals. Let

$$\Sigma = \mathbb{R} \times [0, 1], \quad \partial\Sigma = \mathbb{R} \times \{0, 1\}.$$

For $\beta > 0$, define $G_\beta : \Sigma \rightarrow \Sigma$ by

$$G_\beta(x, y) = (x + \beta, 1 - y).$$

Let G_β^k denote the k -fold composition of G_β or G_β^{-1} , according as $k > 0$ or $k < 0$, and let G_β^0 be the identity. A *clean triangle* is the convex hull of three non-collinear points in $\partial\Sigma$. A *clean triangulation* of Σ is a countable, locally finite collection of clean triangles with pairwise disjoint interiors and union Σ . It is G_β -invariant if G_β preserves the triangulation. A *squeeze map* of a clean triangle is an affine map into $\mathbb{R}^3$ which decreases the length of the edge contained in $\partial\Sigma$ and increases the lengths of the other two edges. A number $\beta > 0$ is *realized* if there is a continuous map $f : \Sigma \rightarrow \mathbb{R}^3$ such that

1. f restricts to a squeeze map on each triangle of a G_β -invariant clean triangulation of Σ ;
2. $f(p) = f(q)$ if and only if $p = G_\beta^k(q)$ for some $k \in \mathbb{Z}$.

Then the infimum of the set of realized numbers is $\sqrt{3}$.

Theorem 2 (Established stronger claim). *The realized numbers are exactly the interval $(\sqrt{3}, \infty)$.*

Overview / Strategy

The lower bound passes to the quotient $M_\beta = \Sigma / \langle G_\beta \rangle$, a flat Möbius band with boundary length 2β . The map descends to an embedded piecewise-affine surface. Inside each clean triangle the cevians from the apex to the boundary edge are strictly lengthened. These cevians form a leaf graph; in the quotient the leaf graph is a circle, while upstairs it is a line, and the deck transformation reverses the orientation of the image support line. A topological lemma about paths of oriented lines then forces

two image support lines to be perpendicular and intersecting. Boundary shortening and a Euclidean estimate give $\beta > \sqrt{3}$.

For the upper bound we use one classical external input: the Sadowsky–Wunderlich construction of embedded developable Möbius bands, in a quantitative form. We approximate such a smooth ruled band by an invariant piecewise-linear ruled band whose boundary edges are not expanded and whose transverse edges are shortened by an arbitrarily small amount. A final anisotropic rescaling makes the boundary edges strictly shorter and all transverse edges strictly longer, yielding a realization for every $\beta > \sqrt{3}$.

3.1 The lower bound

Throughout this section assume that β is realized by f on a G_β -invariant clean triangulation $\mathcal{T}$. Write $G = G_\beta$.

Lemma 3 (Face-to-face structure). *The triangulation is face-to-face: distinct edges have disjoint relative interiors, every transverse edge is incident to exactly two triangles, and every boundary edge is incident to exactly one triangle.*

Proof. A clean triangle has exactly one boundary edge: among its three boundary vertices, exactly two lie on one boundary component and the third lies on the other. The remaining two edges are transverse.

Two transverse edges cannot cross in the interior of Σ , since near a transverse crossing each of the two incident triangles occupies one open side of its edge, and two such open half-disks always overlap in an open sector. If two transverse edges are collinear and overlap in more than one point, then they lie on the same non-horizontal line; such a line meets $y = 0$ and $y = 1$ once each, so the two edges are the same geometric segment. Boundary edges cannot overlap in a nontrivial interval, because at an interior point of the overlap both adjacent triangles contain an inward half-neighborhood of the boundary. Finally, the endpoint of a transverse edge cannot lie in the relative interior of a boundary edge: the triangle with the boundary edge contains a small inward half-disk there, while the triangle with the transverse edge contains a nonempty sector inside that half-disk. Hence distinct triangles meet only in common faces.

Let e be a transverse edge and z an interior point of e . A small disk around z is divided by e into two half-disks. At most one triangle occupies each side, by disjointness of interiors, and at least one triangle occupies each side, because the triangles cover Σ . Thus e is incident to exactly two triangles. The same half-disk argument at an interior point of a boundary edge gives exactly one incident triangle. $\square$

Lemma 4 (The finite quotient triangulation). *The quotient map f descends to an embedding $F : M_\beta \rightarrow \mathbb{R}^3$, where*

$$M_\beta = \Sigma / \langle G \rangle = [0, \beta] \times [0, 1] / (0, y) \sim (\beta, 1 - y),$$

and the quotient triangulation of M_β is finite and face-to-face.

Proof. Since $p = Gq$ implies $f(p) = f(q)$, the map descends to a continuous F . If $F([p]) = F([q])$, then $f(p) = f(q)$, so by realization $p = G^k q$, hence $[p] = [q]$. Thus F is injective; because M_β is compact and $\mathbb{R}^3$ is Hausdorff, F is an embedding.

The restriction of f to each triangle is injective. Indeed, if an affine restriction had rank < 2 , then arbitrarily close distinct points p, q in that triangle would satisfy $f(p) = f(q)$. Then $p = G^k q$ for some $k \neq 0$, impossible for sufficiently close p, q , since G^{2n} translates by $2n\beta$ and G^{2n+1} changes the x -coordinate by $(2n+1)\beta$, so every nontrivial G^k moves every point by distance at least β .

Therefore no closed triangle meets a nontrivial translate of itself. If $z \in \tau \cap G^k \tau$, write $z = G^k w$ with $w \in \tau$. Then $f(z) = f(w)$, and injectivity on τ gives $z = w$, contradicting freeness of the G -action unless $k = 0$. A compact fundamental rectangle, for instance $[0, 2\beta] \times [0, 1]$, meets only finitely many closed triangles by local finiteness and compactness, and every orbit of triangles has a representative meeting it. Hence the quotient triangulation is finite. Lemma 3 descends to the quotient. $\square$

Reviewer 2

(Minor comment) The solution gives a rather lengthy verification of the properties stated in these lemmas, which appear to be the intended geometric properties indicated directly by the problem statement. In my view, it would be acceptable to state these properties without proof, or with only a brief justification.

Lemma 5 (Cevian lengthening). *Let $\Delta = \text{conv}\{a, r_0, r_1\}$ be a clean triangle whose boundary edge is $[r_0, r_1]$, and let $A : \Delta \rightarrow \mathbb{R}^3$ be a squeeze map. For $r(s) = (1-s)r_0 + sr_1$, every cevian $[a, r(s)]$ is strictly lengthened:*

$$|A(a) - A(r(s))| > |a - r(s)|, \quad 0 \leq s \leq 1.$$

Proof. Set $q(s) = |A(a) - A(r(s))|^2 - |a - r(s)|^2$. The two transverse edges are strictly lengthened, so $q(0) > 0$ and $q(1) > 0$. Since A is affine,

$$q''(s) = 2(|A(r_1) - A(r_0)|^2 - |r_1 - r_0|^2) < 0$$

because the boundary edge is strictly shortened. Thus q is strictly concave and lies above its chord: $q(s) \geq (1-s)q(0) + sq(1) > 0$. Taking square roots gives the claim. $\square$

Reviewer 2

(Minor comment) I did not check the computation of $q''(s)$ in detail. However, the conclusion $q''(s) < 0$ is correct. In any case, Lemma 5 is true and admits a routine proof by determining whether $q(s)$ opens upward or downward, which is essentially the strategy implemented in the solution.

Reviewer 2

It is noteworthy that all three of the other AI models tested stated the same lemma as Lemma 2 and introduced the same quadratic function q . Moreover, two of these three models showed that q opens downward by computing q'' .

Lemma 6 (Paths of oriented lines). *Let L_s , $0 \leq s \leq 1$, be a continuous path of oriented lines in $\mathbb{R}^3$. Suppose L_1 is the same unoriented line as L_0 , but with the opposite orientation. Then two distinct parameters give unoriented lines that are perpendicular and intersect.*

Proof. Represent an oriented line by (u, m) , where $u \in S^2$ is its unit direction and $m = p \times u$ for any point p on the line. Reversing orientation sends (u, m) to $(-u, -m)$. Choose continuous $(u(s), m(s))$ and extend by $(u(s+1), m(s+1)) = -(u(s), m(s))$. For $s \in \mathbb{R}/\mathbb{Z}$ and $0 \leq \delta \leq 1$, define

$$a(s, \delta) = u(s) \cdot u(s + \delta), \quad b(s, \delta) = u(s) \cdot m(s + \delta) + u(s + \delta) \cdot m(s).$$

Then $(a, b)(s, 0) = (1, 0)$, $(a, b)(s, 1) = (-1, 0)$, and $(a, b)(s + \delta, 1 - \delta) = -(a, b)(s, \delta)$.

Assume $(a, b) \neq (0, 0)$ everywhere. Then $\Phi = (a, b)/\sqrt{a^2 + b^2} : S^1 \times [0, 1] \rightarrow S^1$ is continuous and is constant on $\delta = 0$. On $\delta = 1/2$,

$$\Phi(s + 1/2, 1/2) = -\Phi(s, 1/2).$$

If $\psi : S^1 \rightarrow S^1$ has $\psi(s + 1/2) = -\psi(s)$, then, after lifting to an angle θ , the integer $(\theta(s + 1/2) - \theta(s) - \pi)/(2\pi)$ is constant, so the degree of ψ is odd. But $\Phi|_{\delta=1/2}$ is homotopic through $S^1 \times [0, 1/2]$ to the constant map $\Phi|_{\delta=0}$, so its degree is zero, a contradiction. Hence $(a, b) = (0, 0)$ somewhere, with $0 < \delta < 1$, so the two parameters are distinct.

At such a zero, $u \cdot v = 0$ for the two directions u, v . If the lines are represented by $(u, p \times u)$ and $(v, q \times v)$, then

$$b = u \cdot (q \times v) + v \cdot (p \times u) = (p - q) \cdot (u \times v).$$

Thus, when $u \perp v$, the condition $b = 0$ says the two nonparallel lines are coplanar. Nonparallel coplanar lines intersect. $\square$

Lemma 7 (Perpendicular intersecting leaves). *In M_β there are two distinct leaves whose image support lines under F are perpendicular and intersect.*

Proof. Inside each triangle take the cevians from the apex to the boundary edge and call them leaves. In the finite quotient triangulation, form a graph Λ_M by taking one closed interval for the leaves of each triangle and gluing endpoints corresponding to common transverse edges. By Lemma 3, every vertex has valence 2.

The graph is connected. If C is a component and $U_C \subset M_\beta$ is the union of the corresponding leaves, then U_C is open: this is immediate inside triangles, along boundary edges, and across transverse edges because the two incident leaf intervals are glued. At a boundary vertex, a small half-disk meets finitely many incident triangle sectors; consecutive sectors share an edge from the vertex, and all non-boundary shared edges are transverse, so all incident triangles lie in the same component. Thus the sets U_C are disjoint open subsets covering the connected surface M_β ; hence there is only one. A finite connected graph with all vertices of valence 2 is a circle, so $\Lambda_M \cong S^1$.

The same openness argument upstairs gives a connected leaf graph Λ_Σ . The action of G on leaves is free. A lifted leaf has endpoints $(x, 0)$ and $(x + t, 1)$. If G^{2n} fixes it, then $2n\beta = 0$. If G^{2n+1} fixes it, then the two endpoint equations give simultaneously $t = (2n + 1)\beta$ and $t = -(2n + 1)\beta$, impossible. Hence $\Lambda_\Sigma \rightarrow \Lambda_M$ is a connected infinite covering of a circle, so $\Lambda_\Sigma \cong \mathbb{R}$.

Orient every lifted leaf from $y = 0$ to $y = 1$. By Lemma 5, its image is a nondegenerate segment, hence determines a continuous oriented support line along Λ_Σ . If ℓ is a leaf, then $G\ell$, reoriented

from bottom to top, has the same image segment with the opposite orientation, because G swaps boundary components and $f \circ G = f$. Applying Lemma 6 to the compact interval in Λ_Σ from ℓ to $G\ell$ gives two distinct quotient leaves with perpendicular intersecting support lines. $\square$

Proposition 8 (No $\beta \leq \sqrt{3}$ is realized). *If β is realized, then $\beta > \sqrt{3}$.*

Proof. Let J be a lifted leaf with endpoints $(x, 0)$ and $(x + t, 1)$. Its flat length is $\sqrt{1 + t^2}$. We first show

$$|t| < \beta. \quad (1)$$

If $t = \beta$ or $t = -\beta$, the endpoints are identified by G or G^{-1} , so their images coincide, contradicting Lemma 5. If $t > \beta$, translate so J has endpoints $(0, 0)$, $(t, 1)$. Then GJ has endpoints $(\beta, 1)$, $(t + \beta, 0)$, and the two segments meet at height $(t + \beta)/(2t) \in (0, 1)$. Thus some distinct $p, q \in J$ satisfy $p = Gq$, so $f(p) = f(q)$, contradicting injectivity on the nondegenerate image segment. The case $t < -\beta$ is identical using G^{-1} .

Parametrize the boundary circle of M_β modulo 2β by assigning coordinate u to $(u, 0)$ and coordinate $v - \beta$ to $(v, 1)$. Hence the boundary arcs between the endpoints of a leaf of span t have lengths

$$\beta - t, \quad \beta + t, \quad (2)$$

both positive by (1). Also every nonconstant boundary arc is strictly shortened by F : the quotient boundary is subdivided into finitely many boundary edges, and on each such edge F has constant length factor $\lambda_e < 1$; summing over subsegments gives

$$\ell_F(\gamma) < \ell_0(\gamma). \quad (3)$$

Reviewer 2

(Minor comment) The symbol γ in Equation (3) is not defined before it is used. Its meaning is therefore somewhat unclear until one reads the later parts of this section.

Choose the two leaves T, B from Lemma 7, and let their support lines meet perpendicularly at P . Since distinct leaves have disjoint relative interiors and F is embedded, P cannot lie in the relative interiors of both image segments. Rename so that $F(B)$ lies in a closed ray from P . Let q be the endpoint of B farther from P , and put $Q = F(q)$. Then

$$y := \text{dist}(Q, \text{line } F(T)) = |Q - P| \geq |F(B)| > |B|_0 \geq 1,$$

so in fact

$$y > 1. \quad (4)$$

Lift T with endpoints $(x, 0)$, $(x + t, 1)$. Replacing this lift by its G -translate changes t to $-t$, so it interchanges the two lengths in (2). Choose the lift so that the boundary arc D containing q has length $\beta + t$; let the other arc be H , of length $\beta - t$. Let A, C be the endpoints of $F(T)$, and set

$$L = |A - C| = |F(T)| > \sqrt{1 + t^2}. \quad (5)$$

The arc H connects A to C , so by (3),

$$\beta - t = \ell_0(H) > \ell_F(H) \geq L > \sqrt{1 + t^2},$$

hence

$$\beta > \sqrt{1+t^2} + t. \quad (6)$$

The arc D connects A to C and passes through Q . Therefore $\ell_F(D) \geq |A-Q| + |Q-C|$. Among points at distance y from the line AC , the sum of distances to A and C is minimized when the projection is the midpoint of AC , giving

$$|A-Q| + |Q-C| \geq \sqrt{L^2 + 4y^2}.$$

Using (4) and (5),

$$\ell_F(D) > \sqrt{5+t^2}.$$

By (3),

$$\beta + t = \ell_0(D) > \ell_F(D) > \sqrt{5+t^2},$$

so

$$\beta > \sqrt{5+t^2} - t. \quad (7)$$

Combining (6) and (7),

$$\beta > \max\{\sqrt{1+t^2} + t, \sqrt{5+t^2} - t\}.$$

For every real t , this maximum is at least $\sqrt{3}$. Indeed, if both terms were $< \sqrt{3}$, then squaring $\sqrt{1+t^2} + t < \sqrt{3}$ gives $t < 1/\sqrt{3}$, while squaring $\sqrt{5+t^2} - t < \sqrt{3}$ gives $t > 1/\sqrt{3}$, a contradiction. Hence $\beta > \sqrt{3}$. $\square$

Reviewer 2

(Minor comment) The proof of the above Lemma appears to implement essentially the same strategies as those given or outlined in Schwartz's paper for similar lemmas. The solution should therefore cite Schwartz's work explicitly in this part of the argument, rather than presenting these proof strategies as if they were novel.

3.2 The upper bound

We quote the classical Sadowsky–Wunderlich construction in the precise form needed.

Theorem 9 (Sadowsky–Wunderlich input). *For every $\mu > \sqrt{3}$, there is a map $E_\mu : \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}^3$ such that:*

1. $E_\mu(X + \mu, 1 - Y) = E_\mu(X, Y)$;
2. *the induced quotient Möbius band is embedded;*
3. *the quotient surface is $C^{1,1}$, piecewise C^∞ , and developable;*
4. *it is foliated by full-width straight rulings, and E_μ maps each ruling affinely and isometrically to a segment;*
5. *the boundary lifts are parametrized by arclength in the X -coordinate;*

6. *the construction remains embedded on a slightly wider strip.*

Proof. This is the three-cylinder construction of Sadowsky, with Wunderlich's analysis of the developable Möbius band; see [1, 2]. In that construction the band consists of three planar pieces and three cylindrical pieces joined along generators with matching tangent planes, and the bending is isometric. The standard separation computation gives maximum half-width

$$b_{\max} = \frac{\mu}{2\sqrt{3}} - \frac{3\pi d}{4}$$

for auxiliary cylinder diameter $d > 0$. Since $\mu > \sqrt{3}$, choosing $0 < d < 2(\mu - \sqrt{3})/(3\pi\sqrt{3})$ gives $b_{\max} > 1/2$, hence a width-one embedded strip, and the strict inequality gives the slightly wider embedded extension. $\square$

For $L, c > 0$, write

$$H_{L,c}(X, Y) = (X + L, c - Y) \quad \text{on } \mathbb{R} \times [0, c].$$

Lemma 10 (Invariant PL approximation). *Let $E : \mathbb{R} \times [0, c] \rightarrow \mathbb{R}^3$ be a scaled Sadowsky band invariant under $H_{L,c}$, with full-width rulings*

$$R_x = [(x, 0), (h(x), c)].$$

Fix x_0 , put $\phi(x) = h(x) + L$, and set

$$S = \max_{x \in [x_0, \phi(x_0)]} |h(x) - x|.$$

For every $D > S$ and every $\varepsilon > 0$, there are an $H_{L,c}$ -invariant clean triangulation $\mathcal{T}_0$ and an $H_{L,c}$ -invariant PL map F_0 such that:

1. $F_0(P) = F_0(Q)$ iff $P = H_{L,c}^k(Q)$ for some $k \in \mathbb{Z}$;
2. every boundary edge is nonexpanded;
3. every transverse edge PQ satisfies

$$|F_0(P) - F_0(Q)| \geq |P - Q| - \varepsilon;$$

4. every transverse edge has horizontal span $< D$.

Proof. The noncrossing full-width rulings imply that h is a strictly increasing homeomorphism. The map $H = H_{L,c}$ sends R_x to $R_{\phi(x)}$, so

$$h(\phi(x)) = x + L, \quad \phi^2(x) = x + 2L. \tag{8}$$

Also

$$|h(x) - x| < L. \tag{9}$$

If $h(x) - x = L$, the endpoints of R_x are identified by H , contradicting the isometric nondegenerate image of R_x . If $h(x) - x > L$, then R_x meets $H(R_x)$ in the interior of the strip; hence two distinct points of R_x have the same image under E , again impossible. The case $h(x) - x < -L$ follows using H^{-1} . Thus $\phi(x) > x$.

Choose $\rho > 0$ with $S + \rho < D$ and $2\rho < \varepsilon$. Choose a partition

$$x_0 < x_1 < \cdots < x_N = \phi(x_0)$$

so fine that $x_{i+1} - x_i < \rho$, $h(x_{i+1}) - h(x_i) < \rho$, all seams of the piecewise smooth Sadowsky band occur among the rulings R_{x_i} , and the embeddedness condition proved below holds. Extend by $x_{i+N} = \phi(x_i)$. Then $x_{i+2N} = x_i + 2L$. Let

$$B_i = (x_i, 0), \quad T_i = (h(x_i), c).$$

The quadrilaterals $B_i B_{i+1} T_{i+1} T_i$ tile the strip. For $0 \leq i < N$, triangulate by the diagonal $B_i T_{i+1}$, and extend H -invariantly. Define F_0 on vertices by $F_0(B_i) = E(B_i)$, $F_0(T_i) = E(T_i)$, and extend affinely on triangles. Then F_0 is H -invariant.

Boundary edges are nonexpanded because their images are chords of boundary arcs parametrized by arclength. A ruling edge is mapped isometrically. For a diagonal $B_i T_{i+1}$, set $d = |B_i - T_{i+1}|$ and $r = |B_i - T_i|$. Since $|T_i - T_{i+1}| < \rho$, one has $|d - r| < \rho$, and

$$|F_0(B_i) - F_0(T_{i+1})| \geq |E(B_i) - E(T_i)| - |E(T_i) - E(T_{i+1})| > r - \rho > d - 2\rho > d - \varepsilon.$$

The other diagonal type is identical, using $x_{i+1} - x_i < \rho$. By (8), the absolute span $|h(x) - x|$ is preserved under ϕ , so all ruling spans are at most S ; diagonal spans are $< S + \rho < D$.

It remains to justify the promised embeddedness for sufficiently fine partition. Let $N = (\mathbb{R} \times [0, c]) / \langle H \rangle$. By Theorem 9, E extends to a slightly wider compact embedded $C^{1,1}$ surface $\widehat{N}$, so there is a tubular neighborhood U with nearest-point projection $\pi : U \rightarrow \widehat{E}(\widehat{N})$. The secant PL maps converge uniformly to E ; because E is affine on rulings, $C^{1,1}$ on each piece, and the seams are included among the ruling edges with matching tangent planes, the maps

$$\Psi = \widehat{E}^{-1} \circ \pi \circ F_0 : N \rightarrow \widehat{N}$$

converge to the inclusion $N \hookrightarrow \widehat{N}$ in C^1 on each piece.

Reviewer 2

(Major comment) First, the map $\widehat{E}$ is not defined before its first appearance in the phrase “nearest-point projection $\pi : U \rightarrow \widehat{E}(\widehat{N})$.” I suspect that the solution intends $\widehat{E}$ to denote the embedding of the wider Moebius strip guaranteed by statement 6 of Theorem 9, but this should be made explicit. Second, the solution claims that the map Φ converges to the inclusion $N \hookrightarrow \widehat{N}$ in C^1 . However, I suspect that Φ itself is only PL, rather than C^1 , so it is unclear what kind of convergence is intended here. This point should be clarified.

For sufficiently small C^1 -error, Ψ is injective. Choose finitely many coordinate disks with convex coordinate images and $\delta > 0$ so that points of N at distance $< \delta$ lie in a common disk. In such a disk, if the derivative of Ψ differs from the identity by $< 1/2$, then integration along the

coordinate segment gives $|\Psi(p) - \Psi(q)| \geq \frac{1}{2}|p - q|$. For pairs with distance $\geq \delta$, compactness and injectivity of the inclusion give a positive minimum separation; sufficiently small C^0 -error preserves this separation. Hence Ψ is injective. If the quotient map induced by F_0 identified two points, then Ψ would identify them, contradiction. Thus the quotient PL map is embedded, which is equivalent to assertion (1) after lifting. $\square$

Proposition 11 (Every $\beta > \sqrt{3}$ is realized). *For every $\beta > \sqrt{3}$, there is a realization of β .*

Proof. Choose μ with $\sqrt{3} < \mu < \beta$, and take the Sadowsky band E_μ . Let $S_\mu = \max |h(x) - x|$ over one fundamental interval of rulings, and choose $D_\mu > S_\mu$. Choose $a < 1$, close to 1, with $a > \mu/\beta$, and set

$$c = \frac{a\beta}{\mu}, \quad L = a\beta.$$

Then $c > 1$ and $L = c\mu$. Scale E_μ by c , giving a width- c , length- L band invariant under $H_{L,c}$. Its span bound scales to cS_μ .

Choose a close enough to 1 so that

$$m := \min_{0 \leq u \leq D_\mu} \left(c\sqrt{1+u^2} - \sqrt{\left(\frac{cu}{a}\right)^2 + 1} \right) > 0. \quad (10)$$

This is possible because at $a = 1$ one has $c = \beta/\mu > 1$, and $c\sqrt{1+u^2} > \sqrt{c^2u^2 + 1}$ on the compact interval $[0, D_\mu]$. Choose $0 < \varepsilon < m$ and apply Lemma 10 to the scaled band with $D = cD_\mu$. We get $\mathcal{T}_0$ and F_0 .

Define

$$A : \Sigma \rightarrow \mathbb{R} \times [0, c], \quad A(x, y) = (ax, cy).$$

Because $L = a\beta$,

$$A(G_\beta(x, y)) = (ax + a\beta, c - cy) = H_{L,c}(A(x, y)).$$

Thus A conjugates G_β to $H_{L,c}$. Put

$$f = F_0 \circ A, \quad \mathcal{T} = A^{-1}(\mathcal{T}_0).$$

Then $\mathcal{T}$ is a countable locally finite G_β -invariant clean triangulation, and f is affine on every triangle.

If pq is a boundary edge, then

$$|f(p) - f(q)| \leq |Ap - Aq| = a|p - q| < |p - q|.$$

If pq is a transverse edge, let s be the horizontal span of $Ap Aq$. By Lemma 10, $s < cD_\mu$; write $s = cu$ with $0 \leq u < D_\mu$. Then

$$|Ap - Aq| = c\sqrt{1+u^2}, \quad |p - q| = \sqrt{\left(\frac{cu}{a}\right)^2 + 1}.$$

Using Lemma 10 and $\varepsilon < m$, (10) gives

$$|f(p) - f(q)| \geq |Ap - Aq| - \varepsilon > |p - q|.$$

Thus f is a squeeze map on every triangle. Finally, for all $p, q \in \Sigma$,

$$f(p) = f(q) \iff F_0(Ap) = F_0(Aq) \iff Ap = H_{L,c}^k(Aq) \text{ for some } k \in \mathbb{Z} \iff p = G_\beta^k(q) \text{ for some } k \in \mathbb{Z}.$$

Therefore β is realized. □

Proof of Theorem 2. Proposition 8 shows that every realized number is $> \sqrt{3}$. Proposition 11 shows that every number $> \sqrt{3}$ is realized. Hence the realized numbers are exactly $(\sqrt{3}, \infty)$. □

Proof of Theorem 1. By Theorem 2, the set of realized numbers is $(\sqrt{3}, \infty)$, whose infimum is $\sqrt{3}$. □

References

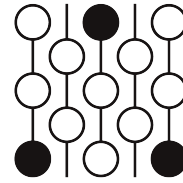
- [1] M. Sadowsky, “Ein elementarer Beweis für die Existenz eines abwickelbaren Möbiusschen Bandes und die Zurückführung des geometrischen Problems auf ein Variationsproblem,” *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse*, 1930, 412–415.
- [2] W. Wunderlich, “Über ein abwickelbares Möbiusband,” *Monatshefte für Mathematik* **66** (1962), 276–289.

Review of Submission 2B

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

Correctness. This solution is completely correct. For the lower bound, it takes an overarching strategy mirroring Schwartz’s proof from his Möbius bands paper, and his model solution. Many of the details are filled in differently than in Schwartz’s solution, for example the proof of cevian lengthening is quite different. A notable exception to this pattern is Lemma 8; see the Novelty section.

For the upper bound, this solution takes the known realizations in the smooth case, finely triangulates them, and then nudges things around by epsilons and deltas to ensure the result is a squeeze map. This part of the argument is executed quite elegantly.

It may also be worth noting that this solution spends a long time at the beginning fussing over certain details that other solutions dispense with much more succinctly.

Novelty. This novelty report shares structure with the novelty report for the other correct solution I refereed. There is no canonical place to put the information common to both, so common information is repeated.

I want to discuss three lemmas where correct solutions varied, and how this solution compares to the others. Then there is the citation issue.

All correct solutions showed that squeeze maps lengthen cevians to the distinguished vertex. All three AI solutions did this by writing down the change in length function and noting that it is a concave-down parabola. The human solution, on the other hand, showed that squeeze maps decrease the distinguished angle, and got cevian length increase from that.

To prove the T -pattern lemma, one can set things up in the right way to reduce to Borsuk-Ulam, or one can directly prove a Borsuk-Ulam-like statement for a slightly different setup. This solution, the human solution, and old versions of Schwartz’s Möbius bands paper opt for the latter approach. The other two correct solutions and the updated version of Schwartz’s Möbius bands paper reduce directly to Borsuk-Ulam.

For the realization half, there were two main philosophies. The first was to construct an explicit triangulation, write out the coordinates for all the faces in terms of ϵ , and show everything works out. The other philosophy was to closely approximate a smooth realization. This solution was one of two correct AI solutions that took the latter approach.

Attribution. The overarching approach by this model is to adapt Schwartz’s Möbius bands paper to a different category. Despite this, the solution never once mentions, alludes to, or cites Schwartz’s work. Lemma 8, which one might call the “optimization” or “geometry” part of the problem, is particularly egregious in its plagiarism. As discussed in the inline comments, it borrows a suspicious amount of notation, the naming for which is motivated by the orientation of a figure featured in Schwartz’s paper. In particular, you would not expect an independent solution, even by someone familiar with the work, to exhibit this common notation.

While the proof is entirely correct, and so will formally receive a “minor revisions” recommendation, a submission of this work by a human to a journal could only be interpreted as willful plagiarism, and would ideally result in a blacklisting.

2 Recommendation

Formally, in accordance with the rubric, this solution receives a “publishable with minor revisions” recommendation, because the only error is a failure of attribution, and this would be easily corrected by citing Schwartz’s prior work.

That said, in the real world this sort of failure of attribution is unacceptable, and would be entirely disqualifying.

3 Submission with inline comments

Theorem 1 (Headline theorem: original problem). Let $\mathbb{Z}$ denote the integers and let $\mathbb{R}$ denote the reals. Let

$$\Sigma = \mathbb{R} \times [0, 1], \quad \partial\Sigma = \mathbb{R} \times \{0, 1\}.$$

For $\beta > 0$, define $G_\beta : \Sigma \rightarrow \Sigma$ by

$$G_\beta(x, y) = (x + \beta, 1 - y).$$

Let G_β^k denote the k -fold composition of G_β or G_β^{-1} , according as $k > 0$ or $k < 0$, and let G_β^0 be the identity. A *clean triangle* is the convex hull of three non-collinear points in $\partial\Sigma$. A *clean triangulation* of Σ is a countable, locally finite collection of clean triangles with pairwise disjoint interiors and union Σ . It is *G_β -invariant* if G_β preserves the triangulation. A *squeeze map* of a clean triangle is an affine map into $\mathbb{R}^3$ which decreases the length of the edge contained in $\partial\Sigma$ and increases the lengths of the other two edges. A number $\beta > 0$ is *realized* if there is a continuous map $f : \Sigma \rightarrow \mathbb{R}^3$ such that

1. f restricts to a squeeze map on each triangle of a G_β -invariant clean triangulation of Σ ;
2. $f(p) = f(q)$ if and only if $p = G_\beta^k(q)$ for some $k \in \mathbb{Z}$.

Then the infimum of the set of realized numbers is $\sqrt{3}$.

Theorem 2 (Established stronger claim). *The realized numbers are exactly the interval $(\sqrt{3}, \infty)$.*

Overview / Strategy

The lower bound passes to the quotient $M_\beta = \Sigma / \langle G_\beta \rangle$, a flat Möbius band with boundary length 2β . The map descends to an embedded piecewise-affine surface. Inside each clean triangle the cevians from the apex to the boundary edge are strictly lengthened. These cevians form a leaf graph; in the quotient the leaf graph is a circle, while upstairs it is a line, and the deck transformation reverses the orientation of the image support line. A topological lemma about paths of oriented lines then forces two image support lines to be perpendicular and intersecting. Boundary shortening and a Euclidean estimate give $\beta > \sqrt{3}$.

Reviewer 3

The term “leaf graph” does not seem to be standard, at least from my googling. The collection of all cevians certainly does not form a graph. When we get to the corresponding section, the AI clarifies what it means: the vertices of the “leaf graph” are the vertices of M_β , and the edges are the edges of ∂M_β . With these definitions, the outlined strategy is correct.

For the upper bound we use one classical external input: the Sadowsky–Wunderlich construction of embedded developable Möbius bands, in a quantitative form. We approximate such a smooth ruled band by an invariant piecewise-linear ruled band whose boundary edges are not expanded and whose transverse edges are shortened by an arbitrarily small amount. A final anisotropic rescaling makes the boundary edges strictly shorter and all transverse edges strictly longer, yielding a realization for every $\beta > \sqrt{3}$.

3.1 The lower bound

Throughout this section assume that β is realized by f on a G_β -invariant clean triangulation $\mathcal{T}$. Write $G = G_\beta$.

Lemma 3 (Face-to-face structure). *The triangulation is face-to-face: distinct edges have disjoint relative interiors, every transverse edge is incident to exactly two triangles, and every boundary edge is incident to exactly one triangle.*

Proof. A clean triangle has exactly one boundary edge: among its three boundary vertices, exactly two lie on one boundary component and the third lies on the other. The remaining two edges are transverse.

Two transverse edges cannot cross in the interior of Σ , since near a transverse crossing each of the two incident triangles occupies one open side of its edge, and two such open half-disks always overlap in an open sector. If two transverse edges are collinear and overlap in more than one point, then they lie on the same non-horizontal line; such a line meets $y = 0$ and $y = 1$ once each, so the two edges are the same geometric segment. Boundary edges cannot overlap in a nontrivial interval, because at an interior point of the overlap both adjacent triangles contain an inward half-neighborhood of the boundary. Finally, the endpoint of a transverse edge cannot lie in the relative interior of a boundary edge: the triangle with the boundary edge contains a small inward half-disk there, while the triangle with the transverse edge contains a nonempty sector inside that half-disk. Hence distinct triangles meet only in common faces.

Let e be a transverse edge and z an interior point of e . A small disk around z is divided by e into two half-disks. At most one triangle occupies each side, by disjointness of interiors, and at least one triangle occupies each side, because the triangles cover Σ . Thus e is incident to exactly two triangles. The same half-disk argument at an interior point of a boundary edge gives exactly one incident triangle. $\square$

Reviewer 3

The lemma statement is certainly correct; the proof seems to be, but is difficult to parse.

Lemma 4 (The finite quotient triangulation). *The quotient map f descends to an embedding $F : M_\beta \rightarrow \mathbb{R}^3$, where*

$$M_\beta = \Sigma / \langle G \rangle = [0, \beta] \times [0, 1] / (0, y) \sim (\beta, 1 - y),$$

and the quotient triangulation of M_β is finite and face-to-face.

Proof. Since $p = Gq$ implies $f(p) = f(q)$, the map descends to a continuous F . If $F([p]) = F([q])$, then $f(p) = f(q)$, so by realization $p = G^k q$, hence $[p] = [q]$. Thus F is injective; because M_β is compact and $\mathbb{R}^3$ is Hausdorff, F is an embedding.

The restriction of f to each triangle is injective. Indeed, if an affine restriction had rank < 2 , then arbitrarily close distinct points p, q in that triangle would satisfy $f(p) = f(q)$. Then $p = G^k q$ for some $k \neq 0$, impossible for sufficiently close p, q , since G^{2n} translates by $2n\beta$ and G^{2n+1} changes the x -coordinate by $(2n+1)\beta$, so every nontrivial G^k moves every point by distance at least β .

Therefore no closed triangle meets a nontrivial translate of itself. If $z \in \tau \cap G^k \tau$, write $z = G^k w$ with $w \in \tau$. Then $f(z) = f(w)$, and injectivity on τ gives $z = w$, contradicting freeness of the G -action unless $k = 0$. A compact fundamental rectangle, for instance $[0, 2\beta] \times [0, 1]$, meets only finitely many closed triangles by local finiteness and compactness, and every orbit of triangles has a representative meeting it. Hence the quotient triangulation is finite. Lemma 3 descends to the quotient. $\square$

Lemma 5 (Cevian lengthening). *Let $\Delta = \text{conv}\{a, r_0, r_1\}$ be a clean triangle whose boundary edge is $[r_0, r_1]$, and let $A : \Delta \rightarrow \mathbb{R}^3$ be a squeeze map. For $r(s) = (1-s)r_0 + sr_1$, every cevian $[a, r(s)]$ is strictly lengthened:*

$$|A(a) - A(r(s))| > |a - r(s)|, \quad 0 \leq s \leq 1.$$

Proof. Set $q(s) = |A(a) - A(r(s))|^2 - |a - r(s)|^2$. The two transverse edges are strictly lengthened, so $q(0) > 0$ and $q(1) > 0$. Since A is affine,

$$q''(s) = 2(|A(r_1) - A(r_0)|^2 - |r_1 - r_0|^2) < 0$$

because the boundary edge is strictly shortened. Thus q is strictly concave and lies above its chord: $q(s) \geq (1-s)q(0) + sq(1) > 0$. Taking square roots gives the claim. $\square$

Lemma 6 (Paths of oriented lines). *Let L_s , $0 \leq s \leq 1$, be a continuous path of oriented lines in $\mathbb{R}^3$. Suppose L_1 is the same unoriented line as L_0 , but with the opposite orientation. Then two distinct parameters give unoriented lines that are perpendicular and intersect.*

Proof. Represent an oriented line by (u, m) , where $u \in S^2$ is its unit direction and $m = p \times u$ for any point p on the line. Reversing orientation sends (u, m) to $(-u, -m)$. Choose continuous $(u(s), m(s))$ and extend by $(u(s+1), m(s+1)) = -(u(s), m(s))$. For $s \in \mathbb{R}/\mathbb{Z}$ and $0 \leq \delta \leq 1$, define

$$a(s, \delta) = u(s) \cdot u(s + \delta), \quad b(s, \delta) = u(s) \cdot m(s + \delta) + u(s + \delta) \cdot m(s).$$

Then $(a, b)(s, 0) = (1, 0)$, $(a, b)(s, 1) = (-1, 0)$, and $(a, b)(s + \delta, 1 - \delta) = -(a, b)(s, \delta)$.

Assume $(a, b) \neq (0, 0)$ everywhere. Then $\Phi = (a, b)/\sqrt{a^2 + b^2} : S^1 \times [0, 1] \rightarrow S^1$ is continuous and is constant on $\delta = 0$. On $\delta = 1/2$,

$$\Phi(s + 1/2, 1/2) = -\Phi(s, 1/2).$$

If $\psi : S^1 \rightarrow S^1$ has $\psi(s + 1/2) = -\psi(s)$, then, after lifting to an angle θ , the integer $(\theta(s + 1/2) - \theta(s) - \pi)/(2\pi)$ is constant, so the degree of ψ is odd. But $\Phi|_{\delta=1/2}$ is homotopic through $S^1 \times [0, 1/2]$ to the constant map $\Phi|_{\delta=0}$, so its degree is zero, a contradiction. Hence $(a, b) = (0, 0)$ somewhere, with $0 < \delta < 1$, so the two parameters are distinct.

At such a zero, $u \cdot v = 0$ for the two directions u, v . If the lines are represented by $(u, p \times u)$ and $(v, q \times v)$, then

$$b = u \cdot (q \times v) + v \cdot (p \times u) = (p - q) \cdot (u \times v).$$

Thus, when $u \perp v$, the condition $b = 0$ says the two nonparallel lines are coplanar. Nonparallel coplanar lines intersect. $\square$

Reviewer 3

It is not clear to me why the two expressions for b are the same, but in a random explicit example they agree, so I am content to believe it. Perhaps it is clear to you, the reader.

Lemma 7 (Perpendicular intersecting leaves). *In M_β there are two distinct leaves whose image support lines under F are perpendicular and intersect.*

Proof. Inside each triangle take the cevians from the apex to the boundary edge and call them leaves. In the finite quotient triangulation, form a graph Λ_M by taking one closed interval for the leaves of each triangle and gluing endpoints corresponding to common transverse edges. By Lemma 3, every vertex has valence 2.

The graph is connected. If C is a component and $U_C \subset M_\beta$ is the union of the corresponding leaves, then U_C is open: this is immediate inside triangles, along boundary edges, and across transverse edges because the two incident leaf intervals are glued. At a boundary vertex, a small half-disk meets finitely many incident triangle sectors; consecutive sectors share an edge from the vertex, and all non-boundary shared edges are transverse, so all incident triangles lie in the same component. Thus the sets U_C are disjoint open subsets covering the connected surface M_β ; hence there is only one. A finite connected graph with all vertices of valence 2 is a circle, so $\Lambda_M \cong S^1$.

The same openness argument upstairs gives a connected leaf graph Λ_Σ . The action of G on leaves is free. A lifted leaf has endpoints $(x, 0)$ and $(x + t, 1)$. If G^{2n} fixes it, then $2n\beta = 0$. If G^{2n+1}

fixes it, then the two endpoint equations give simultaneously $t = (2n + 1)\beta$ and $t = -(2n + 1)\beta$, impossible. Hence $\Lambda_\Sigma \rightarrow \Lambda_M$ is a connected infinite covering of a circle, so $\Lambda_\Sigma \cong \mathbb{R}$.

Orient every lifted leaf from $y = 0$ to $y = 1$. By Lemma 5, its image is a nondegenerate segment, hence determines a continuous oriented support line along Λ_Σ . If ℓ is a leaf, then $G\ell$, reoriented from bottom to top, has the same image segment with the opposite orientation, because G swaps boundary components and $f \circ G = f$. Applying Lemma 6 to the compact interval in Λ_Σ from ℓ to $G\ell$ gives two distinct quotient leaves with perpendicular intersecting support lines. $\square$

Proposition 8 (No $\beta \leq \sqrt{3}$ is realized). *If β is realized, then $\beta > \sqrt{3}$.*

Proof. Let J be a lifted leaf with endpoints $(x, 0)$ and $(x + t, 1)$. Its flat length is $\sqrt{1 + t^2}$. We first show

$$|t| < \beta. \quad (1)$$

If $t = \beta$ or $t = -\beta$, the endpoints are identified by G or G^{-1} , so their images coincide, contradicting Lemma 5. If $t > \beta$, translate so J has endpoints $(0, 0)$, $(t, 1)$. Then GJ has endpoints $(\beta, 1)$, $(t + \beta, 0)$, and the two segments meet at height $(t + \beta)/(2t) \in (0, 1)$. Thus some distinct $p, q \in J$ satisfy $p = Gq$, so $f(p) = f(q)$, contradicting injectivity on the nondegenerate image segment. The case $t < -\beta$ is identical using G^{-1} .

Parametrize the boundary circle of M_β modulo 2β by assigning coordinate u to $(u, 0)$ and coordinate $v - \beta$ to $(v, 1)$. Hence the boundary arcs between the endpoints of a leaf of span t have lengths

$$\beta - t, \quad \beta + t, \quad (2)$$

both positive by (1). Also every nonconstant boundary arc is strictly shortened by F : the quotient boundary is subdivided into finitely many boundary edges, and on each such edge F has constant length factor $\lambda_e < 1$; summing over subsegments gives

$$\ell_F(\gamma) < \ell_0(\gamma). \quad (3)$$

Choose the two leaves T, B from Lemma 7, and let their support lines meet perpendicularly at P . Since distinct leaves have disjoint relative interiors and F is embedded, P cannot lie in the relative interiors of both image segments. Rename so that $F(B)$ lies in a closed ray from P . Let q be the endpoint of B farther from P , and put $Q = F(q)$. Then

$$y := \text{dist}(Q, \text{line } F(T)) = |Q - P| \geq |F(B)| > |B|_0 \geq 1,$$

so in fact

$$y > 1. \quad (4)$$

Lift T with endpoints $(x, 0)$, $(x + t, 1)$. Replacing this lift by its G -translate changes t to $-t$, so it interchanges the two lengths in (2). Choose the lift so that the boundary arc D containing q has length $\beta + t$; let the other arc be H , of length $\beta - t$. Let A, C be the endpoints of $F(T)$, and set

$$L = |A - C| = |F(T)| > \sqrt{1 + t^2}. \quad (5)$$

The arc H connects A to C , so by (3),

$$\beta - t = \ell_0(H) > \ell_F(H) \geq L > \sqrt{1+t^2},$$

hence

$$\beta > \sqrt{1+t^2} + t. \quad (6)$$

The arc D connects A to C and passes through Q . Therefore $\ell_F(D) \geq |A-Q| + |Q-C|$. Among points at distance y from the line AC , the sum of distances to A and C is minimized when the projection is the midpoint of AC , giving

$$|A-Q| + |Q-C| \geq \sqrt{L^2 + 4y^2}.$$

Using (4) and (5),

$$\ell_F(D) > \sqrt{5+t^2}.$$

By (3),

$$\beta + t = \ell_0(D) > \ell_F(D) > \sqrt{5+t^2},$$

so

$$\beta > \sqrt{5+t^2} - t. \quad (7)$$

Combining (6) and (7),

$$\beta > \max\{\sqrt{1+t^2} + t, \sqrt{5+t^2} - t\}.$$

For every real t , this maximum is at least $\sqrt{3}$. Indeed, if both terms were $< \sqrt{3}$, then squaring $\sqrt{1+t^2} + t < \sqrt{3}$ gives $t < 1/\sqrt{3}$, while squaring $\sqrt{5+t^2} - t < \sqrt{3}$ gives $t > 1/\sqrt{3}$, a contradiction. Hence $\beta > \sqrt{3}$. $\square$

Reviewer 3

This argument exactly mirrors that of Schwartz in his paper. In fact, it mirrors the argument so closely that the letters t, T, B, D, H denote the same things in both the paper and this solution. The notation for a few other vertices, namely q, Q, A , and C , are different in this paper than in Schwartz's. Any human presentation of such a proof without mentioning Schwartz's paper would constitute serious plagiarism.

3.2 The upper bound

We quote the classical Sadowsky–Wunderlich construction in the precise form needed.

Theorem 9 (Sadowsky–Wunderlich input). *For every $\mu > \sqrt{3}$, there is a map $E_\mu : \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}^3$ such that:*

1. $E_\mu(X + \mu, 1 - Y) = E_\mu(X, Y)$;
2. the induced quotient Möbius band is embedded;
3. the quotient surface is $C^{1,1}$, piecewise C^∞ , and developable;

4. it is foliated by full-width straight rulings, and E_μ maps each ruling affinely and isometrically to a segment;
5. the boundary lifts are parametrized by arclength in the X -coordinate;
6. the construction remains embedded on a slightly wider strip.

Proof. This is the three-cylinder construction of Sadowsky, with Wunderlich's analysis of the developable Möbius band; see [1, 2]. In that construction the band consists of three planar pieces and three cylindrical pieces joined along generators with matching tangent planes, and the bending is isometric. The standard separation computation gives maximum half-width

$$b_{\max} = \frac{\mu}{2\sqrt{3}} - \frac{3\pi d}{4}$$

for auxiliary cylinder diameter $d > 0$. Since $\mu > \sqrt{3}$, choosing $0 < d < 2(\mu - \sqrt{3})/(3\pi\sqrt{3})$ gives $b_{\max} > 1/2$, hence a width-one embedded strip, and the strict inequality gives the slightly wider embedded extension. $\square$

For $L, c > 0$, write

$$H_{L,c}(X, Y) = (X + L, c - Y) \quad \text{on } \mathbb{R} \times [0, c].$$

Lemma 10 (Invariant PL approximation). *Let $E : \mathbb{R} \times [0, c] \rightarrow \mathbb{R}^3$ be a scaled Sadowsky band invariant under $H_{L,c}$, with full-width rulings*

$$R_x = [(x, 0), (h(x), c)].$$

Fix x_0 , put $\phi(x) = h(x) + L$, and set

$$S = \max_{x \in [x_0, \phi(x_0)]} |h(x) - x|.$$

For every $D > S$ and every $\varepsilon > 0$, there are an $H_{L,c}$ -invariant clean triangulation $\mathcal{T}_0$ and an $H_{L,c}$ -invariant PL map F_0 such that:

1. $F_0(P) = F_0(Q)$ iff $P = H_{L,c}^k(Q)$ for some $k \in \mathbb{Z}$;
2. every boundary edge is nonexpanded;
3. every transverse edge PQ satisfies

$$|F_0(P) - F_0(Q)| \geq |P - Q| - \varepsilon;$$

4. every transverse edge has horizontal span $< D$.

Proof. The noncrossing full-width rulings imply that h is a strictly increasing homeomorphism. The map $H = H_{L,c}$ sends R_x to $R_{\phi(x)}$, so

$$h(\phi(x)) = x + L, \quad \phi^2(x) = x + 2L. \quad (8)$$

Also

$$|h(x) - x| < L. \quad (9)$$

If $h(x) - x = L$, the endpoints of R_x are identified by H , contradicting the isometric nondegenerate image of R_x . If $h(x) - x > L$, then R_x meets $H(R_x)$ in the interior of the strip; hence two distinct points of R_x have the same image under E , again impossible. The case $h(x) - x < -L$ follows using H^{-1} . Thus $\phi(x) > x$.

Choose $\rho > 0$ with $S + \rho < D$ and $2\rho < \varepsilon$. Choose a partition

$$x_0 < x_1 < \cdots < x_N = \phi(x_0)$$

so fine that $x_{i+1} - x_i < \rho$, $h(x_{i+1}) - h(x_i) < \rho$, all seams of the piecewise smooth Sadowsky band occur among the rulings R_{x_i} , and the embeddedness condition proved below holds. Extend by $x_{i+N} = \phi(x_i)$. Then $x_{i+2N} = x_i + 2L$. Let

$$B_i = (x_i, 0), \quad T_i = (h(x_i), c).$$

The quadrilaterals $B_i B_{i+1} T_{i+1} T_i$ tile the strip. For $0 \leq i < N$, triangulate by the diagonal $B_i T_{i+1}$, and extend H -invariantly. Define F_0 on vertices by $F_0(B_i) = E(B_i)$, $F_0(T_i) = E(T_i)$, and extend affinely on triangles. Then F_0 is H -invariant.

Boundary edges are nonexpanded because their images are chords of boundary arcs parametrized by arclength. A ruling edge is mapped isometrically. For a diagonal $B_i T_{i+1}$, set $d = |B_i - T_{i+1}|$ and $r = |B_i - T_i|$. Since $|T_i - T_{i+1}| < \rho$, one has $|d - r| < \rho$, and

$$|F_0(B_i) - F_0(T_{i+1})| \geq |E(B_i) - E(T_i)| - |E(T_i) - E(T_{i+1})| > r - \rho > d - 2\rho > d - \varepsilon.$$

The other diagonal type is identical, using $x_{i+1} - x_i < \rho$. By (8), the absolute span $|h(x) - x|$ is preserved under ϕ , so all ruling spans are at most S ; diagonal spans are $< S + \rho < D$.

It remains to justify the promised embeddedness for sufficiently fine partition. Let $N = (\mathbb{R} \times [0, c]) / \langle H \rangle$. By Theorem 9, E extends to a slightly wider compact embedded $C^{1,1}$ surface $\widehat{N}$, so there is a tubular neighborhood U with nearest-point projection $\pi : U \rightarrow \widehat{E}(\widehat{N})$. The secant PL maps converge uniformly to E ; because E is affine on rulings, $C^{1,1}$ on each piece, and the seams are included among the ruling edges with matching tangent planes, the maps

$$\Psi = \widehat{E}^{-1} \circ \pi \circ F_0 : N \rightarrow \widehat{N}$$

converge to the inclusion $N \hookrightarrow \widehat{N}$ in C^1 on each piece.

For sufficiently small C^1 -error, Ψ is injective. Choose finitely many coordinate disks with convex coordinate images and $\delta > 0$ so that points of N at distance $< \delta$ lie in a common disk. In such a disk, if the derivative of Ψ differs from the identity by $< 1/2$, then integration along the coordinate segment gives $|\Psi(p) - \Psi(q)| \geq \frac{1}{2}|p - q|$. For pairs with distance $\geq \delta$, compactness and

injectivity of the inclusion give a positive minimum separation; sufficiently small C^0 -error preserves this separation. Hence Ψ is injective. If the quotient map induced by F_0 identified two points, then Ψ would identify them, contradiction. Thus the quotient PL map is embedded, which is equivalent to assertion (1) after lifting. $\square$

Proposition 11 (Every $\beta > \sqrt{3}$ is realized). *For every $\beta > \sqrt{3}$, there is a realization of β .*

Proof. Choose μ with $\sqrt{3} < \mu < \beta$, and take the Sadowsky band E_μ . Let $S_\mu = \max |h(x) - x|$ over one fundamental interval of rulings, and choose $D_\mu > S_\mu$. Choose $a < 1$, close to 1, with $a > \mu/\beta$, and set

$$c = \frac{a\beta}{\mu}, \quad L = a\beta.$$

Then $c > 1$ and $L = c\mu$. Scale E_μ by c , giving a width- c , length- L band invariant under $H_{L,c}$. Its span bound scales to cS_μ .

Choose a close enough to 1 so that

$$m := \min_{0 \leq u \leq D_\mu} \left(c\sqrt{1+u^2} - \sqrt{\left(\frac{cu}{a}\right)^2 + 1} \right) > 0. \quad (10)$$

This is possible because at $a = 1$ one has $c = \beta/\mu > 1$, and $c\sqrt{1+u^2} > \sqrt{c^2u^2 + 1}$ on the compact interval $[0, D_\mu]$. Choose $0 < \varepsilon < m$ and apply Lemma 10 to the scaled band with $D = cD_\mu$. We get $\mathcal{T}_0$ and F_0 .

Define

$$A : \Sigma \rightarrow \mathbb{R} \times [0, c], \quad A(x, y) = (ax, cy).$$

Because $L = a\beta$,

$$A(G_\beta(x, y)) = (ax + a\beta, c - cy) = H_{L,c}(A(x, y)).$$

Thus A conjugates G_β to $H_{L,c}$. Put

$$f = F_0 \circ A, \quad \mathcal{T} = A^{-1}(\mathcal{T}_0).$$

Then $\mathcal{T}$ is a countable locally finite G_β -invariant clean triangulation, and f is affine on every triangle.

If pq is a boundary edge, then

$$|f(p) - f(q)| \leq |Ap - Aq| = a|p - q| < |p - q|.$$

If pq is a transverse edge, let s be the horizontal span of $ApAq$. By Lemma 10, $s < cD_\mu$; write $s = cu$ with $0 \leq u < D_\mu$. Then

$$|Ap - Aq| = c\sqrt{1+u^2}, \quad |p - q| = \sqrt{\left(\frac{cu}{a}\right)^2 + 1}.$$

Using Lemma 10 and $\varepsilon < m$, (10) gives

$$|f(p) - f(q)| \geq |Ap - Aq| - \varepsilon > |p - q|.$$

Thus f is a squeeze map on every triangle. Finally, for all $p, q \in \Sigma$,

$$f(p) = f(q) \iff F_0(Ap) = F_0(Aq) \iff Ap = H_{L,c}^k(Aq) \text{ for some } k \in \mathbb{Z} \iff p = G_\beta^k(q) \text{ for some } k \in \mathbb{Z}.$$

Therefore β is realized. □

Proof of Theorem 2. Proposition 8 shows that every realized number is $> \sqrt{3}$. Proposition 11 shows that every number $> \sqrt{3}$ is realized. Hence the realized numbers are exactly $(\sqrt{3}, \infty)$. □

Proof of Theorem 1. By Theorem 2, the set of realized numbers is $(\sqrt{3}, \infty)$, whose infimum is $\sqrt{3}$. □

References

- [1] M. Sadowsky, “Ein elementarer Beweis für die Existenz eines abwickelbaren Möbiusschen Bandes und die Zurückführung des geometrischen Problems auf ein Variationsproblem,” *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse*, 1930, 412–415.
- [2] W. Wunderlich, “Über ein abwickelbares Möbiusband,” *Monatshefte für Mathematik* **66** (1962), 276–289.